EX. A - 1

ITERATIVE POLICY EVALUATION

IN: A POLICY π TO BE EVALUATED, K

- INITIALIZE $V(s)$ $\forall s \in \mathcal{S}$, EXCEPT $V(s) = 0$ IF s IS TERMINAL
- FOR 1 : K

$$V'(s) \leftarrow V(s)$$

$$V(s) \leftarrow \sum_{a \in A} \pi(a|s) \sum_{s', r} p(s', r | s, a) \left[r + \gamma V'(s') \right]$$

END

2 - 3

$\boxed{K=0}$

V(s)

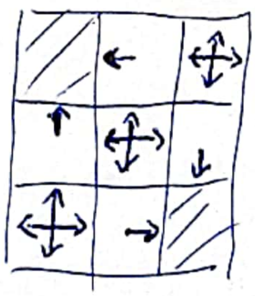
0	0	0
0	0	0
0	0	0

POLICY

$\boxed{K=1}$

0	-1	-1
-1	-1	-1
-1	-1	0

POLICY IMPROV.



$$V(s_2) = V(s_3) = \ldots = V(s_8)$$

$$= \frac{1}{2}(-1 + 0 - 1 + 0) = -1$$

$\boxed{K=2}$

$V(s)$

0	-2	-2
$-3/2$	-2	$-3/2$
-2	-2	0

POLICY IMPROV.

	$\leftarrow$	$\downarrow$
$\uparrow$	$\leftrightarrow$	$\downarrow$
$\uparrow$	$\rightarrow$	

OPTIMAL

K=2 IS ENOUGH

$$V(s_4) = V(s_6) = \frac{1}{2}(-1+0-1-1) = -\frac{3}{2}$$

$$V(s_3) = V(s_2) = V(s_5) = V(s_7) = V(s_8) \overset{!}{=} \frac{1}{2}(-1-1-1-1) = -2$$

$\boxed{K=3}$

$V(s)$

0	-3	$-\frac{11}{4}$
-2	-3	-2
$-\frac{11}{4}$	-3	0

POLICY IMPROV.

	$\leftarrow$	$\downarrow$
$\uparrow$	$\leftrightarrow$	$\downarrow$
$\uparrow$	$\rightarrow$	

OPTIMAL

$$V(s_4) = V(s_6) = \frac{1}{2}(-1+0-1-2) = -2$$

$$V(s_3) = V(s_7) = \frac{1}{2}\left(-1-\frac{3}{2}-1-2\right) = -\frac{11}{4}$$

$$V(s_2) = V(s_5) = V(s_8) = \frac{1}{2}(-1-2-1-2) = -3$$

$\boxed{K=4}$

$V(s)$

0	-4	$-\frac{27}{8}$
$-\frac{19}{8}$	-4	$-\frac{19}{8}$
$-\frac{27}{8}$	-4	0

POLICY IMPROV.

	$\leftarrow$	$\downarrow$
$\uparrow$	$\leftrightarrow$	$\downarrow$
$\uparrow$	$\rightarrow$	

$$V(s_4) = V(s_6) = \frac{1}{2}\left(-1+0-1-\frac{11}{4}\right) = -\frac{19}{8}$$

$$V(s_3) = V(s_7) = \frac{1}{2}\left(-1-2-1-\frac{11}{4}\right) = -\frac{27}{8}$$

$$V(s_2) = V(s_5) = V(s_8) = \frac{1}{2}(-1-3-1-3) = -4$$